

Data visualization

1.Horizontal Bar Plot for

Ozone concentration in air

```
barplot(airquality$Ozone, main = 'Ozone Concentration in air', xlab = 'ozone levels', horiz = TRUE)
```

2.Vertical Bar Plot for

Ozone concentration in air

```
barplot(airquality$Ozone, main = 'Ozone Concentration in air', xlab = 'ozone levels', col='blue', horiz = FALSE)
```

3.Histogram for Maximum Daily Temperature

```
data(airquality)
```

```
hist(airquality$Temp, main = "La Guardia Airport's\ Maximum Temperature(Daily)", xlab = "Temperature(Fahrenheit)", xlim = c(50, 125), col = "yellow", freq = TRUE)
```

4.Box plot for average wind speed

```
data(airquality)
```

```
boxplot(airquality$Wind, main = "Average wind speed\ at La Guardia Airport", xlab = "Miles per hour", ylab = "Wind", col = "orange", border = "brown", horizontal = TRUE, notch = TRUE)
```

5.Multiple Box plots, each representing

an Air Quality Parameter

```
boxplot(airquality[, 0:4], main = 'Box Plots for Air Quality Parameters')
```

6.Scatter plot for Ozone Concentration per month

```
data(airquality)
```

```
plot(airquality$Ozone, airquality$Month, main = "Scatterplot Example", xlab = "Ozone Concentration in parts per billion", ylab = "Month of observation ", pch = 19)
```

7.Set seed for reproducibility

```
data <- matrix(rnorm(50, 0, 5), nrow = 5, ncol = 5) colnames(data) <- paste0("col", 1:5) rownames(data) <- paste0("row", 1:5) heatmap(data)
```

8.print using ggplot

```
ggplot(data = mpg) + geom_point(mapping = aes(x = displ, y = hwy))
```

9.print using aesthetic

```
ggplot(data = mpg) + geom_point(mapping = aes(x = displ, y = hwy, color = class))
```

10.Take the first faceted plot in this section

```
ggplot(data = mpg) + geom_point(mapping = aes(x = displ, y = hwy)) + facet_wrap(~ class, nrow = 2)
```

11.print a point graph

```
ggplot(data = mpg) + geom_point(mapping = aes(x = displ, y = hwy))
```

12.print a smooth curve

```
ggplot(data = mpg) + geom_smooth(mapping = aes(x = displ, y = hwy))
```

13.print different types of curves

```
ggplot(data = mpg) + geom_smooth(mapping = aes(x = displ, y = hwy, linetype = drv))
```

14.display the smooth with diff colour

```
ggplot(data = mpg) + geom_smooth( mapping = aes(x = displ, y = hwy, color = drv), show.legend = FALSE )
```

15.display multiple geoms in the same plot, add multiple geom functions

```
ggplot(data = mpg) + geom_point(mapping = aes(x = displ, y = hwy)) + geom_smooth(mapping = aes(x = displ, y = hwy))
```